

For this task, you must write the **relational algebra operations** for the following queries (your answer must show an understanding of query efficiency)
project: π , select: σ , join: $\bowtie$, intersect: $\cap$, union: $\cup$, minus: $-$

Questions

- a. List the doctor ID, last name and phone number for all doctors specialising in vascular surgery (spec_code = "VS"). **[1 mark]**

$R1 = \pi \text{ doctor_id } (\sigma \text{ spec_code} = \text{"VS"} \text{ DOCTOR_SPECIALITY})$

$R = R1 \bowtie (\pi \text{ doctor_id, doctor_lname, doctor_phone DOCTOR})$

- b. List the procedure code, procedure name and procedure standard cost for all procedures that have not been used during any admission. **[2 marks]**

$R1 = \pi \text{ proc_code, proc_name, proc_std_cost PROCEDURE}$

$R = R1 - (R1 \bowtie \pi \text{ proc_code ADM_PRC})$

- c. List the patient ID, first and last name and contact phone number for all patients who have had the procedure named "Angiogram" carried out during their admission. **[3 marks]**

$R1 = \pi \text{ patient_id, patient_fname, patient_lname, patient_contact_phn PATIENT}$

$R2 = \pi \text{ proc_code } (\sigma \text{ proc_name} = \text{"Angiogram"} \text{ PROCEDURE})$

$R3 = \pi \text{ adm_no, proc_code ADM_PRC } \bowtie R2$

$\pi \text{ patient_id ADMISSION}$

$R4 = R3 \bowtie \pi \text{ adm_no, patient_id ADMISSION}$

$R = R4 \bowtie R1$